Ex.No.3-Scenario-Based Report Development Utilizing Diverse Prompting Techniques for the the following Prompt Engineering types with examples - Straightforward Prompts - Tabular Format Prompting - Missing Word Prompting - Preceding Question Prompting.

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REGISTER NUMBER : 212225240061

Aim: To write the prompts for these following prompt types and evaluate that using any one method

1. Straightforward Prompts

2. Tabular Format Prompting

3. Preceding Question Prompting

4. Missing Word Prompting

Explanation - Any one use case from Unit 5 and generate the report for that with the unit 2 Prompt type

Procedure:

1. Straightforward Prompts:

- "Define photosynthesis in one sentence."

- "List three advantages of electric vehicles."

2. Tabular Format Prompting:

- "Compare and contrast AC and DC current in a table."

- "Provide a table listing five programming languages, their paradigms, and one use case each."

3. Preceding Question Prompting:

- "Why is climate change a global concern? Explain how greenhouse gases contribute to global warming."

- "How do vaccines work? Describe the process of immunization in simple terms."

4. Missing Word Prompting:

- "The capital of France is ____."

- "In photosynthesis, plants absorb sunlight to produce ____."

OUTPUT:

1. Straightforward Prompts:
"Define photosynthesis in one sentence."

- Photosynthesis is the process by which green plants and certain other organisms transform light energy into chemical energy, using water and carbon dioxide to synthesize nutrients—specifically sugars—while releasing oxygen as a byproduct.

"List three advantages of electric vehicles."

- Electric vehicles (EVs) offer significant benefits that range from personal savings to global environmental impact. Here are three primary advantages:

Environmental Impact:
EVs produce zero tailpipe emissions, significantly reducing air pollution and greenhouse gas emissions compared to internal combustion engines.
Lower Operating Costs:
Electricity is generally cheaper than gasoline, and because EVs have fewer moving parts (no oil changes, spark plugs, or exhaust systems), maintenance costs are much lower over the vehicle's lifetime.
Performance and Quietness:
Electric motors provide instant torque, resulting in smooth and rapid acceleration, while also operating much more quietly than traditional engines.

2. Tabular Format Prompting:
   - "Compare and contrast AC and DC current in a table."

Feature	Alternating Current (AC)	Direct Current (DC)
Direction of Flow	Reverses direction periodically.	"Flows in a single, constant direction."
Source	"Power plants (generators), wall outlets."	"Batteries, solar cells, fuel cells."
Long-Distance Travel	Efficient for long distances with low energy loss.	Difficult to transmit over long distances (higher loss).
Frequency	Typically 50Hz or 60Hz depending on the country.	Zero frequency; it is a steady line.
Voltage Change	Easily stepped up or down using a transformer.	Difficult to change voltage levels efficiently.
Common Uses	"Household appliances (fridge, lights), industrial motors."	"Smartphones, laptops, EVs, LED flashlights."

- "Provide a table listing five programming languages, their paradigms, and one use case each."

Language	Primary Paradigm	Common Use Case
Python	"Multi-paradigm (Object-oriented, Procedural)"	Data Science and Machine Learning

JavaScript	"Multi-paradigm (Event-driven, Functional)"	Web Development (Front-end and Back-end)
C++	"Multi-paradigm (Object-oriented, Generic)"	Game Development and High-performance Systems
Haskell	Purely Functional	Academic Research and Financial Modeling
SQL	Declarative	Database Management and Data Querying

3. Preceding Question Prompting:

- "Why is climate change a global concern? Explain how greenhouse gases contribute to global warming."

- The Role of Greenhouse GasesTo understand global warming, it is helpful to look at the Greenhouse Effect.
- While some heat retention is necessary to keep Earth habitable, human activities have increased the concentration of specific gases, trapping excess heat.
- Solar Radiation:
  The sun sends energy to Earth in the form of light. The Earth absorbs some of this energy and radiates the rest back toward space as infrared radiation (heat).
- The "Blanket" Effect: Greenhouse gases (GHGs) like carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) act like a thermal blanket.
- They allow visible light to pass through but absorb the outgoing infrared radiation.
- Re-radiation:
  Instead of escaping into space, these gases re-emit the heat in all directions, including back down to the Earth's surface.

Global Consequences
As the concentration of these gases increases, the global average temperature rises, leading to:
Thermal Expansion: Warmer oceans take up more space, contributing to sea-level rise.
Cryosphere Melting: Glaciers and polar ice caps melt, further raising sea levels and altering ocean currents.
Feedback Loops: For example, as Arctic ice melts, the Earth's albedo (reflectivity) decreases, causing the planet to absorb even more heat.

- "How do vaccines work? Describe the process of immunization in simple terms."

Think of a vaccine as a "training exercise" for your immune system. Instead of waiting for a dangerous disease to strike, a vaccine introduces a safe, weakened, or inactive version of the germ to your body so your "internal security team" can learn how to fight it.

The 4-Step Process of ImmunizationThe goal of immunization is to create immunity without the person actually having to get sick first.
--- Here is how that happens:
- The Introduction (The "Wanted" Poster):
  The vaccine introduces an antigen—a tiny, harmless piece of a virus or bacteria (like a protein or a dead version of the germ). This acts as a "Wanted" poster for your immune system.
- The Recognition:
  Your white blood cells (specifically B-cells and T-cells) detect this foreign invader. They realize it doesn't belong and begin building a defense strategy.
- The Response:

Your body produces antibodies. These are specialized proteins designed to lock onto that specific germ and neutralize it.

- The Memory:

Once the "mock" infection is cleared, your body creates memory cells. These cells stay in your system for years, remembering exactly what that germ looks like and how to kill it.

4. Missing Word Prompting:

- "The capital of France is ____."
  The capital of France is Paris.
- "In photosynthesis, plants absorb sunlight to produce ____."
  In photosynthesis, plants absorb sunlight to produce glucose (sugar) and oxygen.

RESULT:

The various types of Prompts are executed successfully with generated the report.